Table 2.5

p	q	$p \vee q$	$p \rightarrow (p \vee q)$	$\neg p$	$\neg p \wedge q$	$p \wedge (\neg p \wedge q)$
0	0	0	1	1	0	0
0	1	1	1	1	1	0
1	0	1	1	0	0	0
1	1	1	1	0	0	0

Definition 2.1 A compound statement is called a *tautology* if it is true for all truth value assignments for its component statements. If a compound statement is false for all such assignments, then it is called a *contradiction.*

Throughout this chapter we shall use the symbol T_0 to denote any tautology and the symbol F_0 to denote any contradiction.

We can use the ideas of tautology and implication to describe what we mean by a valid argument. This will be of primary interest to us in Section 2.3, and it will help us develop needed skills for proving mathematical theorems. In general, an argument starts with a list of *given* statements called *premises* and a statement called the *conclusion* of the argument. We examine these premises, say $p_1, p_2, p_3, \ldots, p_n$, and try to show that the conclusion q follows logically from these given statements—that is, we try to show that if each of $p_1, p_2, p_3, \ldots, p_n$ is a true statement, then the statement q is also true. To do so one way is to examine the implication

$$(p_1 \wedge p_2 \wedge p_3 \wedge \cdots \wedge p_n)\dagger \rightarrow q,$$

where the hypothesis is the conjunction of the n premises. If any one of $p_1, p_2, p_3, \ldots, p_n$ is false, then no matter what truth value q has, the implication $(p_1 \wedge p_2 \wedge p_3 \wedge \cdots \wedge p_n) \rightarrow q$ is true. Consequently, if we start with the premises $p_1, p_2, p_3, \ldots, p_n$—each with truth value 1—and find that under these circumstances q also has the value 1, then the implication

$$(p_1 \wedge p_2 \wedge p_3 \wedge \cdots \wedge p_n) \rightarrow q$$

is a *tautology* and we have a *valid argument.*

EXERCISES 2.1

1. Determine whether each of the following sentences is a statement.
   a) In 1990 George Bush was the president of the United States.
   b) $x + 3$ is a positive integer.
   c) If only every morning could be as sunny and clear as this one!

† At this point we have dealt only with the conjunction of two statements, so we must point out that the conjunction $p_1 \wedge p_2 \wedge p_3 \wedge \cdots \wedge p_n$ of n statements is true if and only if each p_i, $1 \leq i \leq n$, is true. We shall deal with this generalized conjunction in detail in Example 4.14 of Section 4.2.

d) Fifteen is an even number.
e) If Jennifer is late for the party, then her cousin Zachary will be quite angry.
f) What time is it?
g) From the halls of Montezuma to the shores of Tripoli.
h) As of June 30, 1986, Christine Marie Evert had won the French Open seven times.

2. Identify the primitive statements in Exercise 1.

3. Let p, q be primitive statements for which the implication $p \to q$ is false. Determine the truth values for
a) $p \wedge q$ **b)** $\neg p \vee q$ **c)** $q \to p$ **d)** $\neg q \to \neg p$

4. Let p, q, r, s denote the following statements: p: I finish writing my computer program before lunch; q: I shall play tennis in the afternoon; r: The sun is shining; s: The humidity is low. Write the following in symbolic form.
a) If the sun is shining, I shall play tennis this afternoon.
b) Finishing the writing of my computer program before lunch is necessary for my playing tennis this afternoon.
c) Low humidity and sunshine are sufficient for me to play tennis this afternoon.

5. Let p, q, r denote the following statements about a particular triangle ABC. p: Triangle ABC is isosceles; q: Triangle ABC is equilateral; r: Triangle ABC is equiangular. Translate each of the following into an English sentence.
a) $q \to p$ **b)** $\neg p \to \neg q$ **c)** $q \leftrightarrow r$
d) $p \wedge \neg q$ **e)** $r \to p$

6. Determine the truth value of each of the following implications.
a) If $3 + 4 = 12$, then $3 + 2 = 6$.
b) If $3 + 3 = 6$, then $3 + 6 = 9$.
c) If $3 + 3 = 6$, then $3 + 4 = 9$.
d) If Thomas Jefferson was the third president of the United States, then $2 + 3 = 5$.

7. Rewrite each of the following statements as an implication in the If-Then form.
a) Practicing her serve daily is a sufficient condition for Darci to have a good chance of winning the tennis tournament.
b) Fix my air conditioner or I won't pay the rent.
c) Mary will be allowed on Larry's motorcycle only if she wears her helmet.

8. Construct a truth table for each of the following compound statements, where p, q, r denote primitive statements.
a) $\neg(p \vee \neg q) \to \neg p$ **b)** $p \to (q \to r)$ **c)** $(p \to q) \to r$
d) $(p \to q) \to (q \to p)$ **e)** $[p \wedge (p \to q)] \to q$ **f)** $(p \wedge q) \to p$
g) $q \leftrightarrow (\neg p \vee \neg q)$ **h)** $[(p \to q) \wedge (q \to r)] \to (p \to r)$

9. Which of the compound statements in Exercise 8 are tautologies?

10. Verify that $[p \to (q \to r)] \to [(p \to q) \to (p \to r)]$ is a tautology.

11. **a)** How many rows are needed for the truth table of the compound statement $(p \vee \neg q) \leftrightarrow [(\neg r \wedge s) \to t]$, where p, q, r, s, and t are primitive statements?
b) Let $p_1, p_2, \ldots, p_n$ denote n primitive statements. Let p be a compound statement that contains at least one occurrence each of p_i, for $1 \le i \le n$—and p contains no other primitive statement. How many rows are needed to construct the truth table for p?

12. Determine all truth value assignments, if any, for the primitive statements p, q, r, s, t that make each of the following compound statements false.
a) $[(p \wedge q) \wedge r] \to (s \vee t)$
b) $[p \wedge (q \wedge r)] \to (s \veebar t)$

18. A segment of a Pascal program contains a Repeat-Until loop structured as follows.

```
Repeat
........
........
Until ((x <> 0) and (y>0)) or (not ((w>0) and (t=3)));
```

For each of the following assignments for the variables x, y, w, and t, determine whether or not the loop terminates.

a) $x = 7, y = 2, w = 5, t = 3$ **b)** $x = 0, y = 2, w = -3, t = 3$
c) $x = 0, y = -1, w = 1, t = 3$ **d)** $x = 1, y = -1, w = 1, t = 3$

19. After baking a pie for the two nieces and two nephews who are visiting her, Aunt Nellie leaves the pie on her kitchen table to cool. Then she drives to the mall to close her boutique for the day. Upon her return she finds that someone has eaten one quarter of the pie (and even had the nerve to leave her or his dirty plate next to the remainder of the pie). Since no one was in her house that day—except for the four visitors—Aunt Nellie questions each niece and nephew about who ate the piece of pie. The four "suspects" tell her the following:

Charles: Kelly ate the piece of pie.
Dawn: I did not eat the piece of pie.
Kelly: Tyler ate the pie.
Tyler: Kelly lied when she said I ate the pie.

If only one of these four statements is true and only one of the four committed this heinous crime, who is the vile culprit who Aunt Nellie will have to punish severely?

2.2 Logical Equivalence: The Laws of Logic

In all areas of mathematics we need to know when the entities we are studying are equal or essentially the same. For example, in arithmetic and algebra we know that two nonzero real numbers are equal when they have the same magnitude and algebraic sign. Hence, for two nonzero real numbers x, y, we have $x = y$ if $|x| = |y|$ and $xy > 0$, and conversely (that is, if $x = y$, then $|x| = |y|$ and $xy > 0$). When we deal with triangles in geometry, the notion of congruence arises. Here triangle ABC and triangle DEF are congruent if, for instance, they have equal corresponding sides—that is, the length of side AB = the length of side DE, the length of side BC = the length of side EF, and the length of side CA = the length of side FD.

Our study of logic is often referred to as the *algebra of propositions* (as opposed to the algebra of real numbers). In this algebra we shall use the truth tables of the statements, or propositions, to develop an idea of when two such entities are essentially the same. We begin with an example.

Example 2.7 For primitive statements p and q, Table 2.6 provides the truth tables for the compound statements $\neg p \vee q$ and $p \rightarrow q$. Here we see that the corresponding truth tables for the two statements $\neg p \vee q$ and $p \rightarrow q$ are exactly the same.